Markov Chains (Continued)

Dr Feng Chen[1]

School of Mathematics and Statistics
UNSW Sydney

[1] Slides adapted from those prepared by previous lecturers of the course, Prof Pierre Del Moral and Dr Leung Chan

Total variation norms

Definition

Probability measures μ_1 and μ_2 on some **finite state space** S.
Total variation distance:

$$\begin{aligned} \|\mu_1 - \mu_2\|_{tv} &= \frac{1}{2} \sum_{x \in S} |\mu_1(x) - \mu_2(x)| \\ &= \sum_{\mu_1 \geq \mu_2} (\mu_1(x) - \mu_2(x)) \\ &= 1 - \sum_{x \in S} [\mu_1 \wedge \mu_2](x) \end{aligned}$$

Definition

Infimum measure $\mu_1 \wedge \mu_2$**:**

$$[\mu_1 \wedge \mu_2](x) := \mu_1(x) \wedge \mu_2(x)$$

For a (**not necessarily finite**) state space S,
Total variation distance:

$$\begin{aligned} \|\mu_1 - \mu_2\|_{tv} &= \frac{1}{2}\int \left| \frac{d\mu_1}{d\lambda}(x) - \frac{d\mu_2}{d\lambda}(x) \right| \lambda(dx) \\ &= 1 - \int [\mu_1 \wedge \mu_2](dx) \end{aligned}$$

where $\mu_i(dx) = p_i(x)\ \lambda(dx)$.

$p_i(x) = \frac{d\mu_i}{d\lambda}(x)$ is the density of the measure μ_i w.r.t. to some common reference measure λ.

Infimum measure $\mu_1 \wedge \mu_2$:

$$[\mu_1 \wedge \mu_2](dx) := [p_1(x) \wedge p_2(x)]\ \lambda(dx)$$

When $\mu_i(dx) = p_i(x)dx$,

$$\begin{aligned} \|\mu_1 - \mu_2\|_{tv} &= \frac{1}{2}\int |p_1(x) - p_2(x)| \; dx \\ [\mu_1 \wedge \mu_2](dx) &= [p_1(x) \wedge p_2(x)] \; dx \end{aligned}$$

The choice of the reference measure λ is NOT unique.

When $\lambda(dx) = q(x)\, dx$,

$$\|\mu_1 - \mu_2\|_{tv} = \frac{1}{2}\int \left| \frac{p_1(x)}{q(x)} - \frac{p_2(x)}{q(x)} \right| \; q(x)\, dx$$

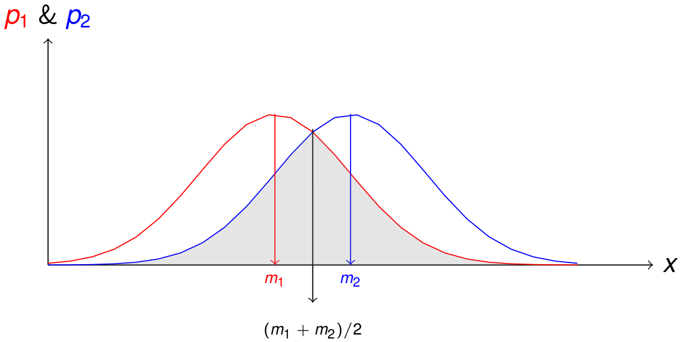
Example 1

Let $Z \sim \mathcal{N}(0,1)$, $X_1 = m_1 + Z$ *and* $X_2 = m_2 + Z$.

$\mu_1 = \mathrm{Law}(X_1)$ *and* $\mu_2 = \mathrm{Law}(X_2)$.

Denote by p_1 *and* p_2 *the Gaussian densities of* μ_1 *and* μ_2.

$$p_1 = \mathcal{N}(m_1, 1), p_2 = \mathcal{N}(m_2, 1)$$



Calculate $\|\mu_1 - \mu_2\|_{tv}$.

Example 2

For two state models, $S = \{0, 1\}$, Calculate $\|\mu_1 - \mu_2\|_{tv}$.

For any function $f \in \mathcal{B}(\{0, 1\})$,

$$\begin{aligned}
\mu_1(f) - \mu_2(f) &= [\mu_1(0)f(0) + (1 - \mu_1(0))f(1)] \\
&\qquad - [\mu_2(0)f(0) + (1 - \mu_2(0))f(1)] \\
&= [\mu_1(0) - \mu_2(0)] \times (f(1) - f(0))
\end{aligned}$$

Definition

For two probability measures μ_1 and μ_2 on the measurable space $(S, \mathcal{S})$, their **total variation distance** is defined by :

$$\|\mu_1 - \mu_2\|_{tv} = \sup_{A \in \mathcal{S}} |\mu_1(A) - \mu_2(A)|. \tag{1}$$

Proposition 1

Let λ be a common reference measure that dominates both the probability measures μ_1 and μ_2, and let $\mathrm{osc}(f) = \inf\{u : \lambda \otimes \lambda\{(x, y) : |f(x) - f(y)| > u\} = 0\}$, *and* $\|f\|_\infty = \inf\{u : \lambda\{x : |f(x)| > u\} = 0\}$. *Then:*

$$\begin{aligned} & \|\mu_1 - \mu_2\|_{tv} \\ = \quad & \sup\{|\mu_1(f) - \mu_2(f)| \;:\; f \text{ s.t. } \mathrm{osc}(f) \le 1\} && (2) \\ = \quad & \frac{1}{2} \sup\{|\mu_1(f) - \mu_2(f)| \;:\; f \text{ s.t. } \|f\|_\infty \le 1\} && (3) \end{aligned}$$

Convergence theorem

Theorem 2

Markov transition $\mathbf{P}$ *on a finite space* S *is* **irreducible** *and* **aperiodic**, *with stationary distribution* π*. Then there exists constant* $\alpha \in (0, 1)$ *and* $C > 0$ *s.t.*

$$\max_{x \in S} \|\mathbf{P}^n(x, \cdot) - \pi\|_{\mathrm{tv}} \leq C\alpha^n$$

Coupling - Total variation distance

Lemma 3

For any couple of r.v. (X, Y), *with* $\mathrm{Law}(X) = \mu_1$ & $\mathrm{Law}(Y) = \mu_2$,

$$\|\mu_1 - \mu_2\|_{tv} \leq \mathbb{P}\,(X \neq Y)$$

Theorem 4

For any probability measures μ_1, μ_2 *on* S,

$$\|\mu_1 - \mu_2\|_{tv} = \inf\left\{\mathbb{P}\,(X \neq Y) \;:\; \mathrm{Law}(X) = \mu_1 \;\&\; \mathrm{Law}(Y) = \mu_2\right\}$$

Definition

The coupling constructed in the proof of Theorem 4 for finite state spaces is called an **maximal coupling** between the distribution $\mathrm{Law}(X) = \mu_1$ and $\mathrm{Law}(Y) = \mu_2$.
In a more simulation based formulation, it is defined by

$$\mathbb{P}\left((X, Y) = (x, y)\right)$$

$$= \alpha_{(\mu_1,\mu_2)} p_{(\mu_1,\mu_2)}(x) 1_x(y) + \left(1 - \alpha_{(\mu_1,\mu_2)}\right) q_{(\mu_1,\mu_2)}(x) q_{(\mu_2,\mu_1)}(y)$$

with the coupling probability $\alpha_{(\mu_1,\mu_2)} = [\mu_1 \wedge \mu_2](S) \in [0, 1[$ and the probability measures

$$p_{(\mu_1,\mu_2)} := \frac{[\mu_1 \wedge \mu_2]}{\alpha(\mu_1, \mu_2)} \quad \text{and} \quad q_{(\mu_1,\mu_2)} := \frac{\mu_1 - [\mu_1 \wedge \mu_2]}{1 - \alpha(\mu_1, \mu_2)}$$

Example 3

(X_1, X_2): Bernoulli $\{0, 1\}$ r.v.'s with parameters $(p_1, p_2) \in [0, 1]^2$ $p_2 \geq p_1$. $\mathrm{Law}(X_i) = \mu_i$, with $i = 1, 2$. Check that

$$\|\mu_1 - \mu_2\|_{tv} = |p_1 - p_2|$$

Example 4

(X_1, X_2): r.v.'s on S with distributions (μ_1, μ_2), U: r.v. on $\mathcal{U}$. Consider a couple of r.v.'s

$$X_1' = F(X_1, U_1) \quad \textit{and} \quad X_2' = F(X_2, U_2)$$

where $F : (\mathcal{U} \times S) \mapsto S'$, and (U_1, U_2) are two (not necessarily independent) copies of U. If μ_1' & μ_2' stand for the distributions of X_1' & X_2' respectively, check that

$$\|\mu_1' - \mu_2'\|_{tv} \leq \|\mu_1 - \mu_2\|_{tv}.$$

Stopping times and coupling

Definition

X_n: Markov chain in some state space S.
Recall that a **stopping time** T is a r.v. taking values in

$\mathbb{N} \cup \{\infty\}$ s.t. the events $\{T = n\}$ depend **only** on the r.v.'s $X_0, \ldots, X_n$.

Example 5

The first time a Markov chain on a finite set S hits some subset A.

$$T = \inf \{n \geq 0 \ : \ X_n \in A\}$$

is a stopping time since $\{T = n\} = \{X_0 \notin A, \ \ldots, X_{n-1} \notin A \, , X_n \in A\}$.

Example 6

T*: the first exit time of the set* $B = S - A$.

Example 7

T': The first time a given chain reaches its maximum $X_n^\star := \max_{0\leq p\leq n} X_p$ on some interval $[0, n]$. by

$$T' = \inf\{n \geq 0 \; : \; X_p = X_n^\star\}$$

T' is not *a stopping time since the event $\{T' = k\}$ depend on the whole sequence of states from the origin up to the terminal time n.*

Example 8

T'': last exit or hitting times of the form

$$T'' = \max\{n \geq 0 \; : \; X_n \in A, A \subset S\}$$

T'' are not *stopping times since the events $\{T'' = n\}$ also depend on the future of the chain.*

$$\{T'' = n\} = \cap_{p\geq 1}\{X_n \in A \; , \; X_{n+p} \notin A\}$$

Proposition 5 (strong Markov property)

For any Markov chain taking values in some state space S, and for any stopping time T,

$$\mathbb{P}\left(X_{n+1} \in dx_{n+1} \mid T = n,\ (X_0, \ldots, X_n) = (x_0, \ldots, x_n)\right)$$

$$= \mathbb{P}\left(X_{n+1} \in dx_{n+1} \mid X_n = x_n\right)$$

Markov chain $(X_{T+n})_{n\geq 0}$ is again a Markov chain with initial condition X_T.

Proposition 6

X_n & Y_n: a couple of Markov chains s.t. $X_n = Y_n$ for any n after some stopping time T. Then, we have

$$\|\mathrm{Law}(X_n) - \mathrm{Law}(Y_n)\|_{tv} \leq \mathbb{P}\left(X_n \neq Y_n\right) \leq \mathbb{P}\left(T > n\right) \leq \frac{\mathbb{E}(T)}{n}$$

In particular,

$$\begin{aligned} \mathbb{P}(T < \infty) = 1 \quad &\Rightarrow \quad \mathbb{P}\left(T > n\right) \rightarrow_{n\uparrow\infty} 0 \\ &\Rightarrow \quad \|\mathrm{Law}(X_n) - \mathrm{Law}(Y_n)\|_{tv} \rightarrow_{n\uparrow\infty} 0 \end{aligned}$$

Strong stationary times

Definition

X_n: Markov chain taking values in some state space S, with an invariant measure π.
A **strong stationary time** T (a.k.a **strong uniform time**) is a **stopping time** s.t. X_T and T are independent and $\mathrm{Law}(X_T) = \pi$.

Theorem 7

For any strong uniform time T,

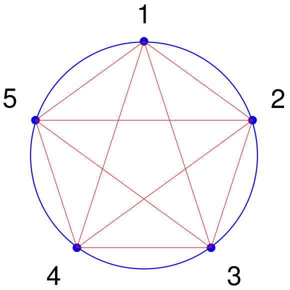
$$\|\mathrm{Law}(X_n) - \pi\|_{tv} \leq \mathbb{P}(T > n)$$

Example 9 (Markov chains on complete graphs)

Markov chain with transition $\mathbf{P}(x, y) = 1/d$ *on a finite and complete graph with* d *vertices* $S := \{1, \ldots, d\}$.

This chain is reversible with the unique invariant uniform measure $\pi(x) = 1/d, \forall\, x \in S$.

When $d = 5$,



Let (X_n, Y_n): *independent Markov chain;* T: *the first time they meet. Show that:*

1. T *is a* **geometric r.v.** *with success parameter* $1/d$.
2. $\mathbb{P}(T > n) \leq e^{-n/d}$.

Coupling

Markov chains with pre-specified invariant distributions

- Metropolis-Hastings algorithm
- Gibbs sampler

Markov Chain Monte Carlo methods

Purpose: Design and run a Markov chain $X_n, n = 0, 1, \ldots,$ in state space S with its invariant distribution equal to a prespecified target distribution π on S, so we can approximate the measure $\pi(\cdot)$ by the occupation measures $\frac{1}{N}\sum_{i\leq N-1}\delta_{X_i}(\cdot)$, and $\pi(f)$ by the sample means $\frac{1}{N}\sum_{i\leq N-1} f(X_i)$ for N large.

Metropolis-Hastings (M-H) algorithm

Choose a transition kernel $K(x, \mathrm{d}y)$ such that $\pi(\cdot) \ll K(x, \cdot)$ for all $x \in S$. Set

$$G(x, y) = \frac{\pi(\mathrm{d}y)K(y, \mathrm{d}x)}{\pi(\mathrm{d}x)K(x, \mathrm{d}y)}, \tag{4}$$

with the convention $0/0 = 0$; and

$$a(x, y) = 1 \wedge G(x, y), \text{ or } a(x, y) = \frac{G(x, y)}{1 + G(x, y)}.$$

M-H Algorithm

Starting from an initial value X_0, at time step $n \geq 1$, first generate a Y according to $K(X_{n-1}, \mathrm{d}y)$, and then set X_n to Y with probability $a(X_{n-1}, Y)$, and to X_{n-1} with probability $1 - a(X_{n-1}, Y)$.

Proposition 8

The M-H algorithm leads to a Markov chain with a transition kernel $M(x, \mathrm{d}y)$ given by

$$M(x, \mathrm{d}y) = K(x, \mathrm{d}y)a(x, y) + (1 - \int K(x, \mathrm{d}z)a(x, z))\delta_x(\mathrm{d}y),$$

which is reversible w.r.t. the target measure π; that is,

$$\pi(\mathrm{d}x)M(x, \mathrm{d}y) = \pi(\mathrm{d}y)M(y, \mathrm{d}x).$$

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Gibbs sampler

Let π be a target measure defined on some product space $S = (S_1 \times S_2)$, with the following disintegration property:

$$\pi(\mathrm{d}x_1, \mathrm{d}x_2) = \pi_1(\mathrm{d}x_1)L_{1,2}(x_1, \mathrm{d}x_2) = \pi_2(\mathrm{d}x_2)L_{2,1}(x_2, \mathrm{d}x_1),$$

whether π_1 and π_2 are the two marginal probability measures on S_1 and S_2 respectively, and $L_{1,2}$ and $L_{2,1}$ are the corresponding conditional probability measures.

Gibbs sampler

The Gibbs sampler is the Markov chain with transition kernel

$$M = K_1 K_2,$$

where the K_i, $i = 1, 2$ are given by

$$K_1((x_1, x_2), (\mathrm{d}y_1, \mathrm{d}y_2)) = \delta_{x_1}(\mathrm{d}y_1)L_{1,2}(y_1, \mathrm{d}y_2),$$
$$K_2((x_1, x_2), (\mathrm{d}y_1, \mathrm{d}y_2)) = \delta_{x_2}(\mathrm{d}y_2)L_{2,1}(y_2, \mathrm{d}y_1).$$

Proposition 9

The transitions K_1 and K_2 are reversible w.r.t. the measure π. In addition, the Metropolis Hastings transitions M_1, and respectively M_2, with proposal transition K_1, and respectively K_2, and acceptance rate $a = 1 \wedge G$ with G given by (4) have unit acceptance rate.

Example 10 (Rare event simulation)

Let X_1 *and* X_2 *be independent* $N(0,1)$ *variables. Design a Markov chain for drawing samples from the conditional distribution of* (X_1, X_2) *given* $X_1^2 + X_2^2 \geq 50$*. Check your design by simulating the chain.*